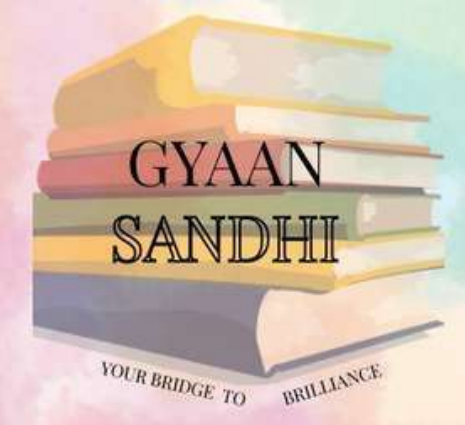
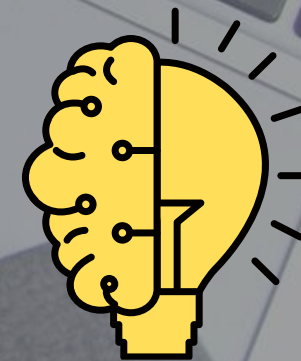




UNIT 3

Reporting Authoring

Business Intelligence



Relational Data Model

Introduction

- The Relational Data Model is a method of organizing data into tables called relations.
- Each table consists of rows and columns.
- Rows represent records, while columns represent attributes or fields of data.
- It allows users to store, retrieve, and manipulate data easily using SQL (Structured Query Language).
- Relationships between tables are maintained using keys such as Primary Key and Foreign Key.
- This model helps maintain data consistency, accuracy, and easy accessibility.



Relational Data Model

Features of Relational Data Model

1. Data Organized in Tables

- Data is stored in the form of tables.
- Each table contains information about a single entity such as Customers, Products, Employees, etc.

2. Primary Key

- A Primary Key is a column or group of columns that uniquely identifies each row in a table.
- It ensures that no duplicate records exist.

Example:

- CustomerID in the Customers table.

3. Foreign Key

- A Foreign Key is a column that creates a relationship between two tables.
- It refers to the Primary Key of another table.

Example:

- CustomerID in the Orders table refers to CustomerID in the Customers table.



Relational Data Model

Example of Relational Data Model

Customers Table

CustomerID	Name	City
101	Rishi	Nashik
102	Darshan	Mumbai

Orders Table

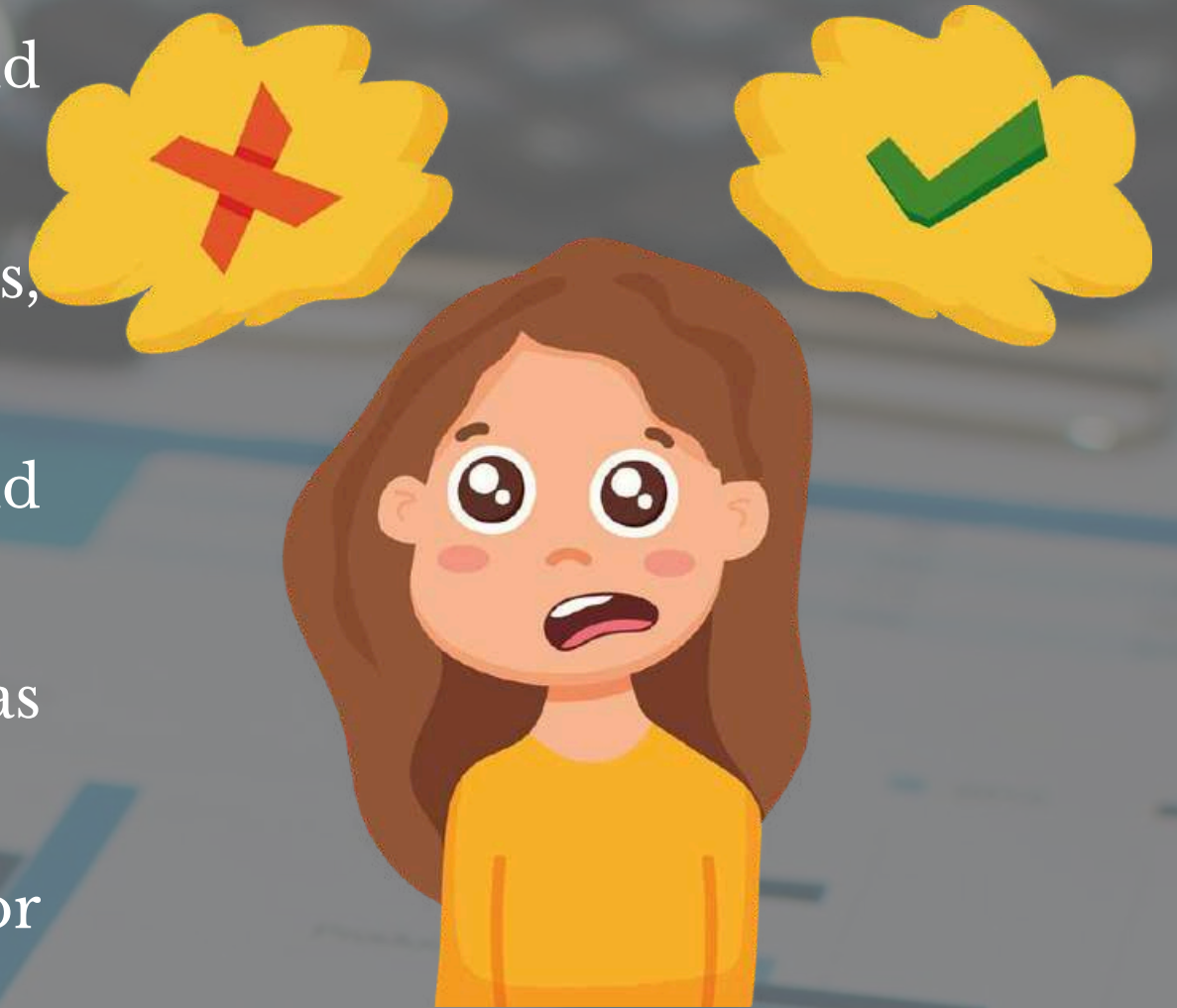
OrderID	CustomerID	Product	Amount
1	101	Laptop	50000
2	102	Headphone	2000



Multidimensional Data Model

Introduction

- The Multidimensional Data Model is mainly used in Data Warehousing and OLAP (Online Analytical Processing) systems.
- Unlike the Relational Data Model, it organizes data into multiple dimensions, similar to a cube structure.
- This structure allows faster and more flexible data analysis, reporting, and decision-making.
- Each dimension in the cube represents a different way to analyze data, such as Time, Product, or Region.
- At the intersection of these dimensions, measures like Sales, Revenue, Profit, or Quantity are stored.



Case Study Example: Retail Superstore

1. Dimensions

Product Dimension

- Category
- Subcategory

Multidimensional Data Model

Time Dimension

- Year
- Quarter
- Month

Region Dimension

- Country
- State
- City

2. Measures

- Sales Amount
- Profit
- Quantity Sold

Sales Cube Structure

- One side of the cube represents Time (2024, 2025, etc.).
- Second side represents Product (Electronics, Furniture, etc.).
- Third side represents Region (Nashik, Pune, etc.).
- Inside the cube, values such as Sales and Profit are stored.

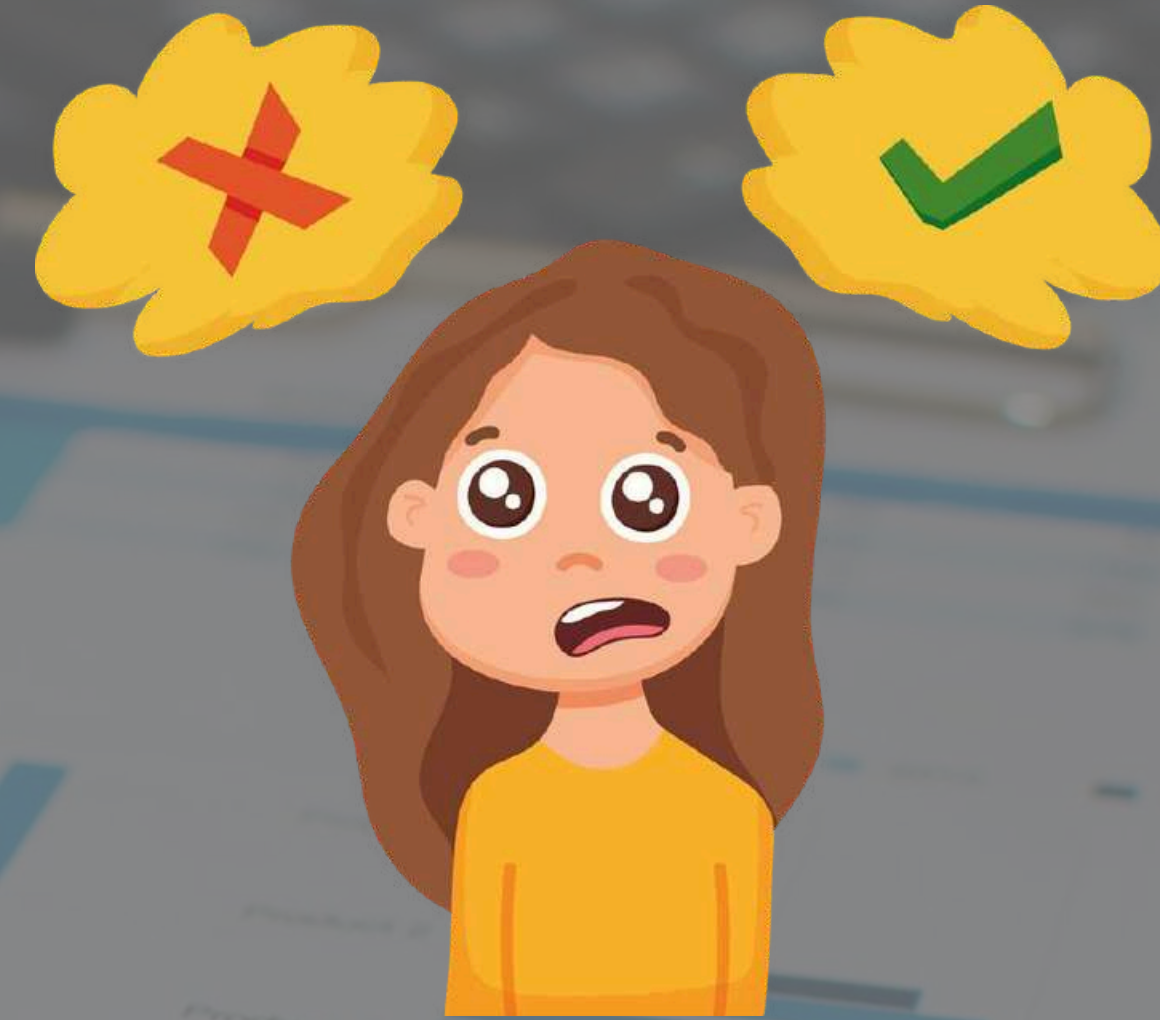
Advantages of Multidimensional Data Model

1. Fast Query Performance

- Data is pre-aggregated in cubes.
- Queries execute much faster compared to relational models.

2. Supports Complex Analytics

- OLAP operations like:



Multidimensional Data Model

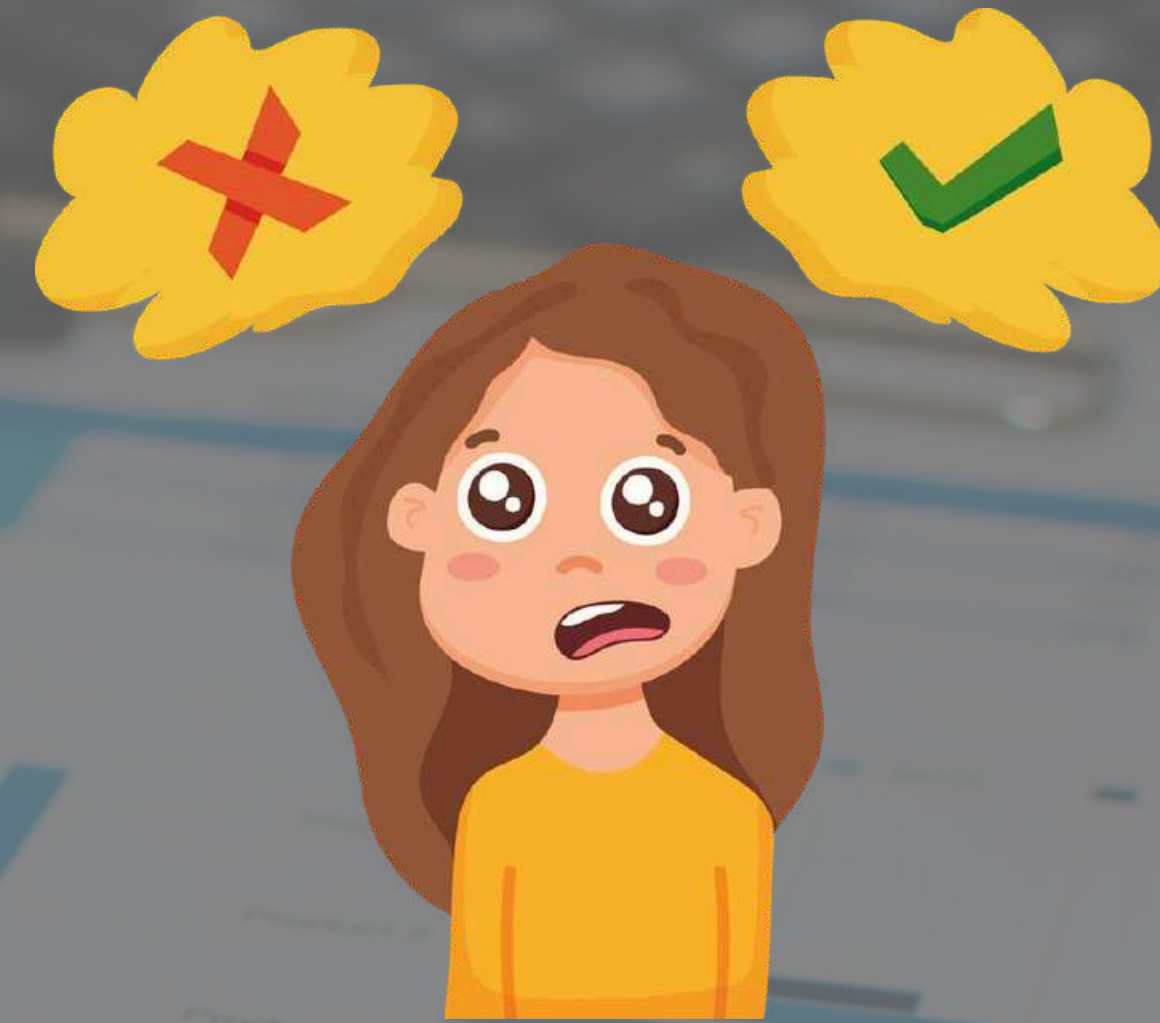
- Slicing
- Dicing
- Drill-down
- Roll-up
- These operations help in deep data analysis.

3. Better Decision Making

- Provides summarized and comparative views of data.
- Helps managers make informed business decisions.

Applications

- Business Intelligence Systems
- Sales Analysis
- Financial Reporting
- Market Analysis
- Data Warehousing Systems



Types of Reports

1. List Report (Tabular Report)

- A List Report displays raw data in a simple row-and-column format.
- It is ideal for viewing detailed records such as transactions, customer details, employee data, etc.
- Each row represents a record, and each column represents an attribute or field.

Example: Customer details table

- Transaction records

2. Cross Tab Report (Pivot Table)

- A Cross Tab Report presents summarized data across two dimensions, such as Region vs Product.
- Totals and aggregates are displayed inside cells.
- It is mainly used in sales and financial analysis to compare metrics across categories.

Example:

- Sales by Region and Product Category



Types of Reports

3. Chart Report

- Chart Reports use visual elements such as:
 - Bar charts
 - Pie charts
 - Line charts
- These reports help users easily understand trends and patterns.
- Commonly used in:
 - Performance dashboards
 - Trend analysis
 - Monthly reviews

Example: A line chart showing monthly revenue growth.

4. Dashboard Report

- A Dashboard Report combines multiple visual elements such as:
 - Charts
 - KPIs



Types of Reports

- Maps
- Tables
- It provides a high-level summary of important metrics in real time.
- Used by managers and executives to monitor business performance and make quick decisions.

Example:

- Business performance dashboard showing sales, profit, and customer growth.

5. Ad-Hoc Report

- An Ad-Hoc Report is created on demand for specific business questions or scenarios.
- Mainly used by analysts and managers for one-time analysis and quick decision-making.

Example:

- Finding how many units of a product were sold during a flash sale in a particular city.



Types of Reports

6. Exception Report

- An Exception Report displays only records that meet specific conditions such as:
 - Errors
 - Missing data
 - Outliers
- Used in:
 - Auditing
 - Compliance monitoring
 - Alert systems
- Helps businesses identify and act on problems quickly.

Example:

- Report showing transactions with unusually high amounts.

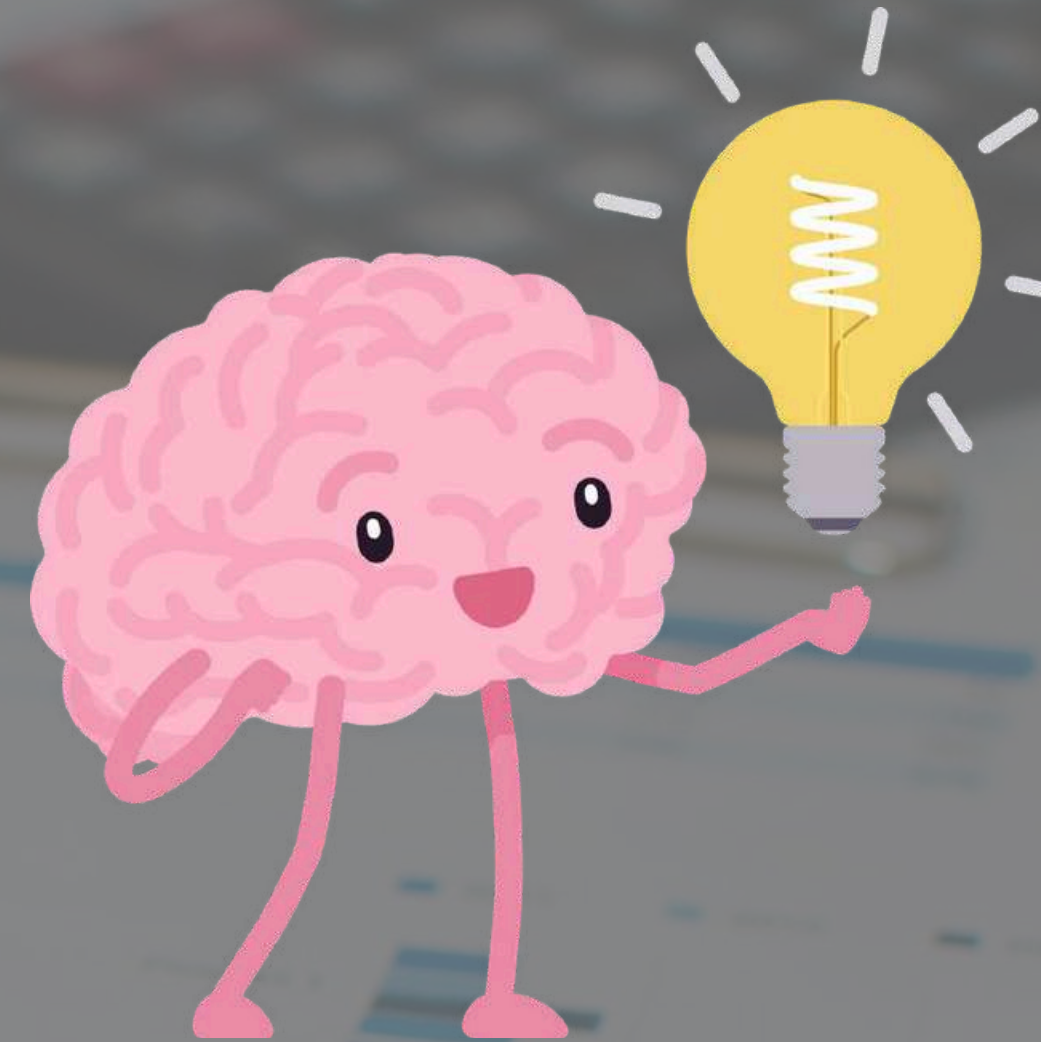


Data Grouping and Data Sorting

Data Grouping

Definition

- Data Grouping means organizing similar data items together based on a common field or category.
- It is the process in Business Intelligence (BI) where records sharing a common value are placed into one group.
- It is commonly used in reports to organize and summarize information.
- Data can be grouped by Region, Product, Date, Department, etc.



Example: Grouping Sales Data by Region

Before Grouping

Before Grouping

Order ID	Region	Amount
101	East	500
102	West	700
103	East	300
104	West	600

Data Grouping and Data Sorting

After Grouping by Region

Region: East

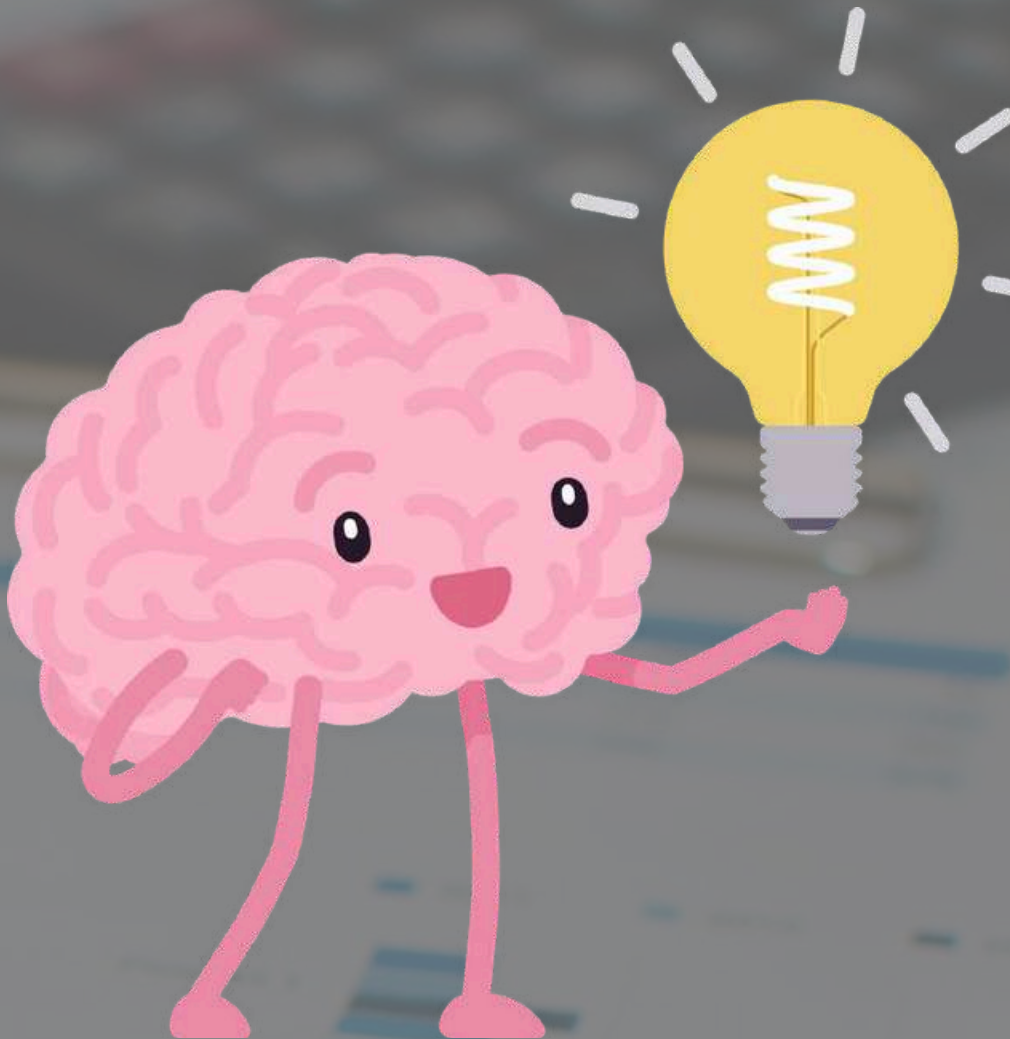
Order ID	Amount
101	500
103	300

Total for East = 800

Region: West

Order ID	Amount
102	700
104	600

Total for West = 1300



Data Grouping and Data Sorting

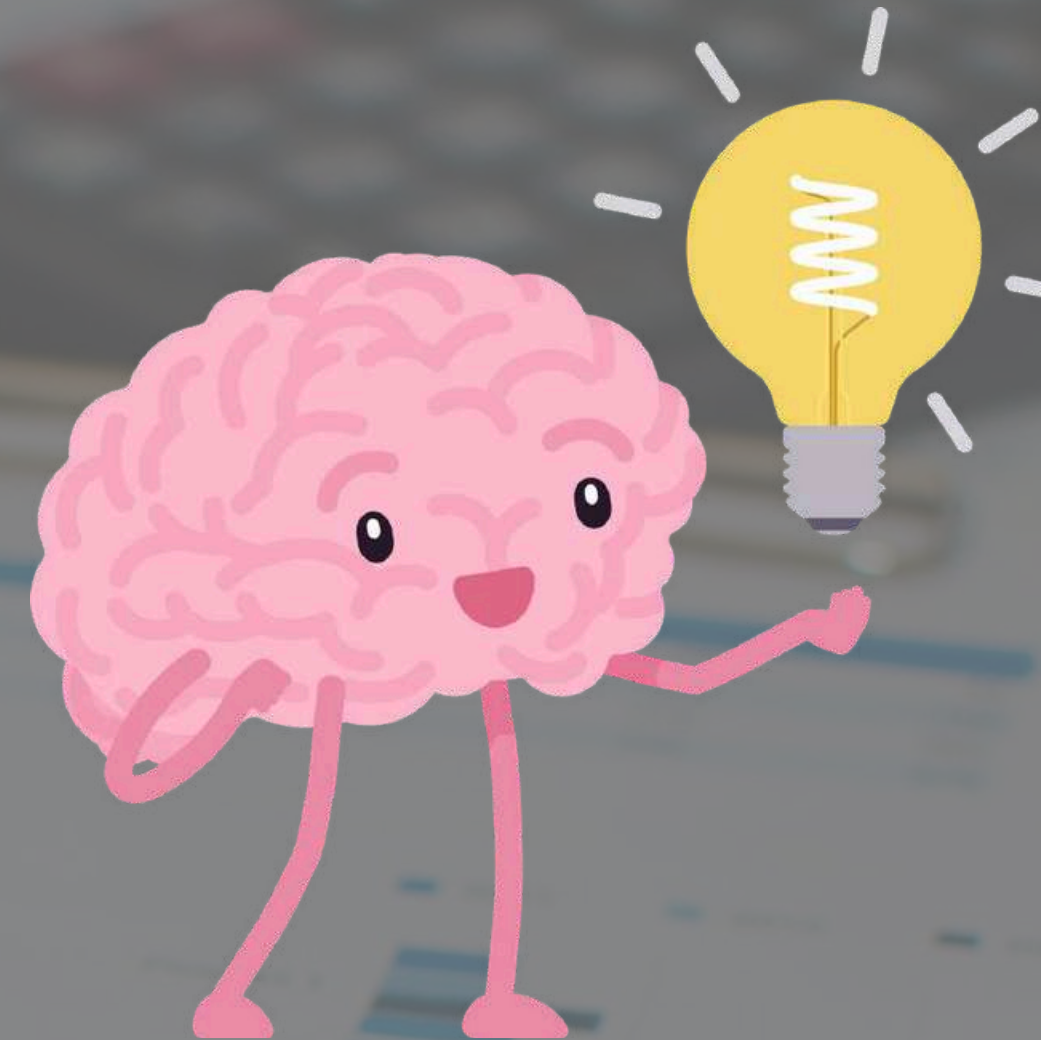
Benefits of Data Grouping

- Improves report organization.
- Makes data easier to understand.
- Provides summarized information.
- Helps in analysis and decision-making.

Data Sorting

Definition

- Data Sorting means arranging data in a specific order based on one or more fields.
- Data can be sorted in:
 - Ascending Order (A–Z, Low to High)
 - Descending Order (Z–A, High to Low)
- Sorting makes data easier to read, compare, and analyze.
- Reports can be sorted by names, dates, prices, sales, and other fields.



Data Grouping and Data Sorting

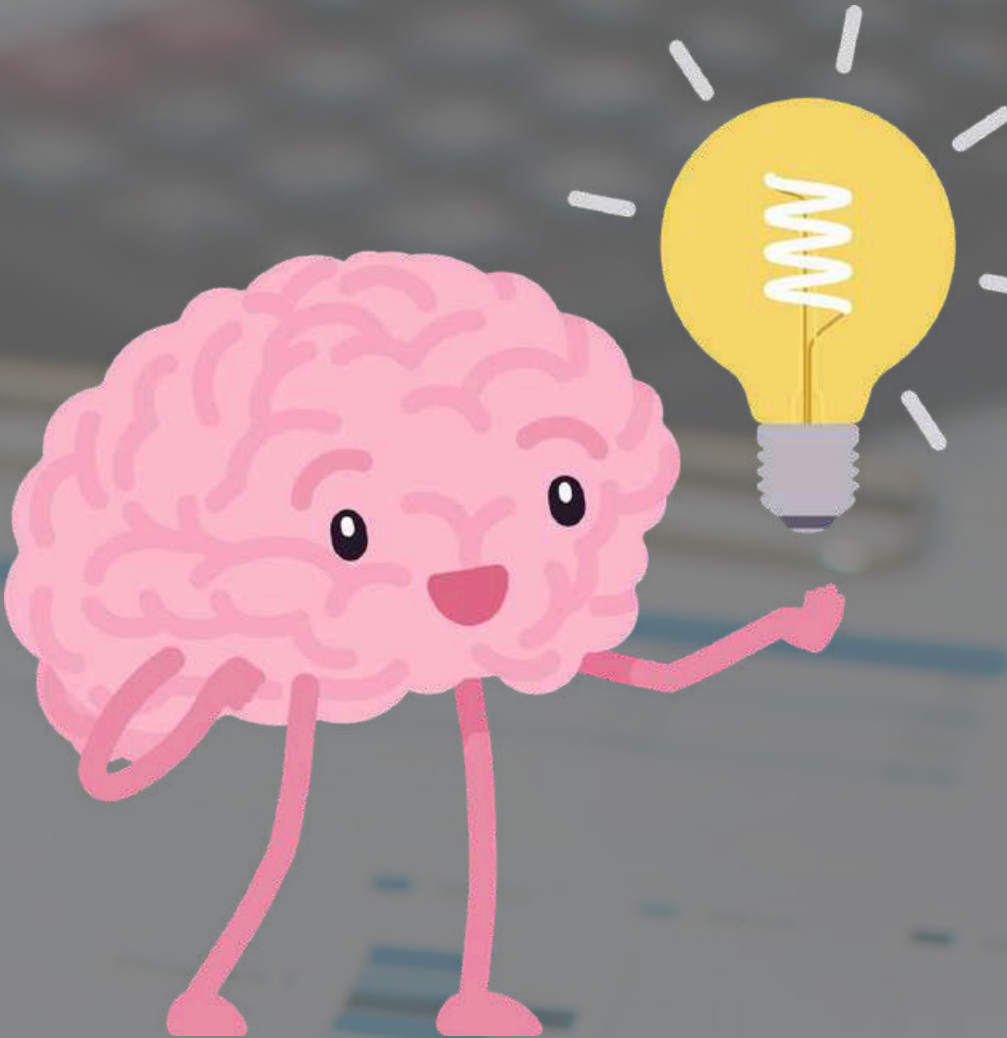
Example: Sorting Products by Price

Before Sorting

Product Name	Price
Mouse	300
Keyboard	500
Monitor	8000
Cable	150

After Sorting (By Price - Ascending)

Product Name	Price
Cable	150
Mouse	300
Keyboard	500
Monitor	8000



Filtering Reports

Definition

- Filtering reports means displaying only the data that meets specific conditions or criteria.
- It is an important feature in Business Intelligence (BI) reporting.
- Filters allow users to limit the data displayed in a report based on predefined rules.
- Instead of showing all records, filtering helps focus only on relevant information.
- Users can filter data by month, region, product, customer, date range, and many other fields.
- Filtering is especially useful in dashboards and ad-hoc reports where users need dynamic views of data.
- Filters can be:
 - Simple Filters: Show sales above ₹500.
 - Complex Filters: Show sales from the last quarter for Region A while excluding Product X.

Example: Sales Report



Filtering Reports

Example: Sales Report

Before Applying Filter

Order ID	Date	Region	Amount
101	2025-03-05	East	500
102	2025-03-15	West	700
103	2025-01-10	East	300

Filter Condition

Show only sales that occurred in March 2025.

After Applying Filter

Order ID	Date	Region	Amount
101	2025-03-05	East	500
102	2025-03-15	West	700



Conditional Formatting and Adding Calculations in Report

Conditional Formatting

Definition

- Conditional Formatting is used to automatically change the appearance of report elements based on specific conditions or rules.
- It can modify:
 - Text color
 - Background color
 - Font style
 - Boldness
 - Icons or indicators
- In Business Intelligence (BI) reports, it helps highlight important data points.
- It makes reports more readable and visually appealing.
- Users can quickly identify trends, issues, and exceptions without reading every value.
- It supports faster analysis and better decision-making.

Example :- Display sales figures in Green if they are above the target.

- Display sales figures in Red if they are below the target.



Conditional Formatting and Adding Calculations in Report

Order Value	Formatting
Above ₹10,000	Green Color
Below ₹5,000	Red Color

Advantages of Conditional Formatting

- Highlights important information instantly.
- Makes reports easier to understand.
- Helps identify problems and opportunities.
- Improves report visualization.
- Supports quick decision-making.



Conditional Formatting and Adding Calculations in Report

Adding Calculations

Definition

- Calculations in reports are used to create new fields or summarize existing data using formulas.
- It is one of the most important features of reporting because it helps users analyze data rather than simply view it.
- Calculated fields provide meaningful business insights.
- Calculations can be used for:
 - Totals
 - Averages
 - Percentages
 - Profit Margins
 - Growth Rates

Uses of Calculations

- Generate new values from existing data.



Conditional Formatting and Adding Calculations in Report

- Measure business performance.
- Analyze trends and efficiency.
- Support data-driven decision-making.

Examples

1. Total Profit :- $\text{Profit} = \text{Sales} - \text{Cost}$

$\text{Profit Margin} = (\text{Profit} / \text{Sales}) \times 100$

Benefits of Adding Calculations

- Converts reports into analytical tools.
- Provides deeper business insights.
- Helps track performance and trends.
- Improves decision-making.
- Reduces manual calculations and errors.



Drill-Down, Drill-Up and Drill-Through

1. Drill-Down

Definition

- Drill-Down is a Business Intelligence (BI) feature that allows users to move from a summarized view of data to a more detailed view.
- It helps users analyze the root cause of trends, variations, or anomalies.
- Users can explore data at different levels of detail.
- Drill-Down is especially useful when data is organized in hierarchies.

Example Hierarchy

- Year → Quarter → Month → Day
- A report may initially show total yearly sales.
- By clicking on a specific year (e.g., 2025), users can view quarterly sales.
- Further drill-down can show monthly sales and then daily sales.

Example



Drill-Down, Drill-Up and Drill-Through

Year	Sales (₹)
2024	1,000,000
2025	1,200,000

Month	Sales (₹)
January	90,000
February	100,000
March	120,000



Drill-Down, Drill-Up and Drill-Through

Advantages of Drill-Down

- Provides detailed analysis.
- Helps identify the root cause of problems.
- Makes reports interactive.
- Avoids overcrowding reports with excessive details.
- Supports better decision-making.

2. Drill-Up

Definition

- Drill-Up is the reverse process of Drill-Down.
- It allows users to move from detailed data back to a summarized or higher-level view.
- Users can navigate through different levels of a hierarchy without reloading the report.

Example Hierarchy

Day → Month → Quarter → Year



Drill-Down, Drill-Up and Drill-Through

Date	Sales (₹)
Mar 1, 2025	4,000
Mar 2, 2025	5,500
Mar 3, 2025	6,000

Daily Sales



Month	Sales (₹)
Mar 2025	120,000

Monthly Sales



Year	Sales (₹)
2025	1,200,000

Yearly Sales



Drill-Down, Drill-Up and Drill-Through

3. Drill-Through

Definition

- Drill-Through is a reporting feature that allows users to move from one report to another while carrying the selected data context.
- It is used when detailed information is stored in a separate report or page.
- It provides deeper analysis without cluttering the main report.

Example

Summary Report

Region	No. of Complaints
North	120
South	90
East	150

Customer Name	Complaint Date	Issue
Rahul Sharma	Mar 5, 2025	Product Defect
Priya Patil	Mar 8, 2025	Late Delivery

Detailed Complaint Report (East Region)



Best Practices in Dashboard Design

Introduction

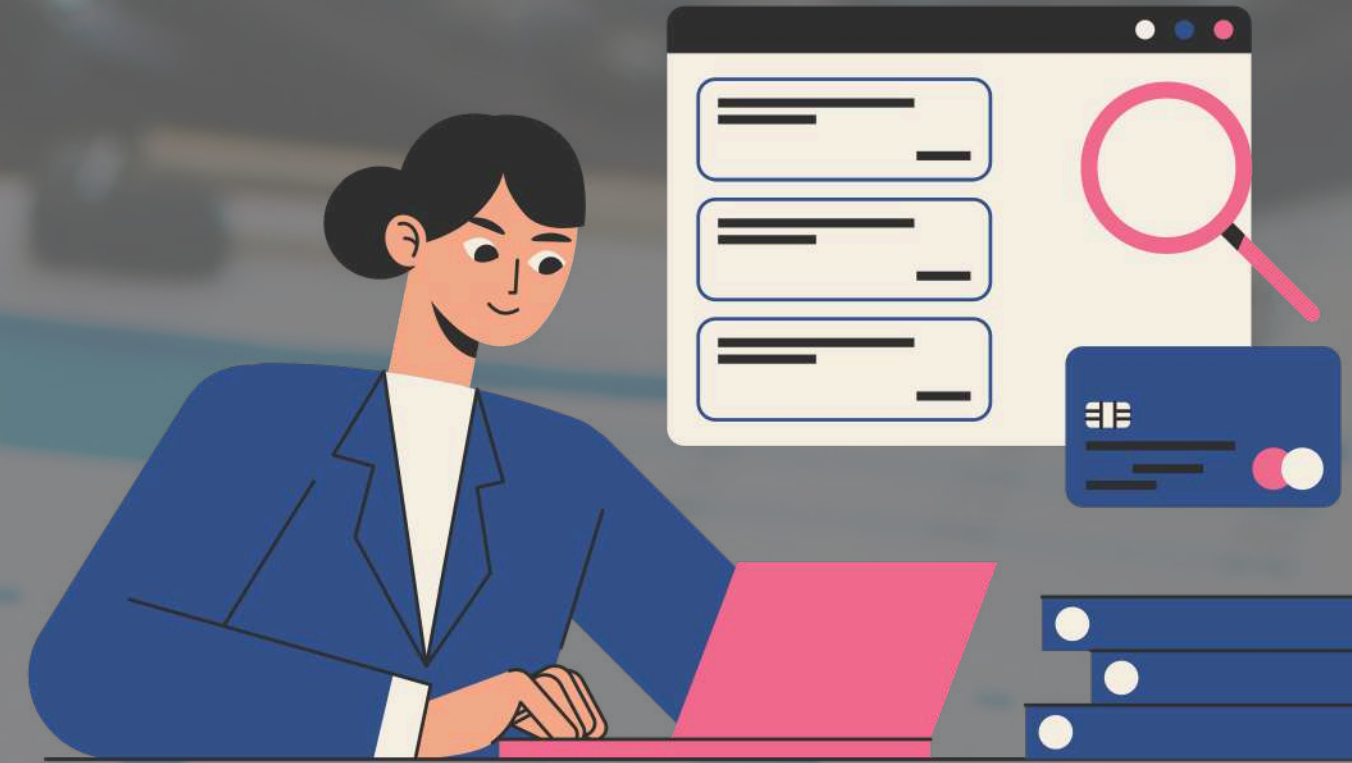
- A dashboard is a visual display of the most important information needed to achieve objectives.
- It presents data on a single screen so that it can be monitored at a glance.
- An effective dashboard helps users identify trends, monitor key metrics, and make informed decisions quickly.

1. Know Your Audience

- Before designing a dashboard, understand who will use it.
- Different users require different types of information.
- A CEO may need a summary of company performance, while a sales manager may require detailed regional sales data.

Example

- CEO: Overall profit, revenue, and growth.
- Sales Manager: Region-wise sales and targets.



Best Practices in Dashboard Design

2. Define Clear Objectives

- Identify the questions the dashboard should answer.
- The dashboard should focus on specific business goals.

Examples

- How are monthly sales trending?
- Which products are underperforming?
- What is the current profit margin?

3. Use the Right Charts for the Right Data

- Selecting the correct chart improves understanding and prevents misinterpretation.
- Different charts serve different purposes.

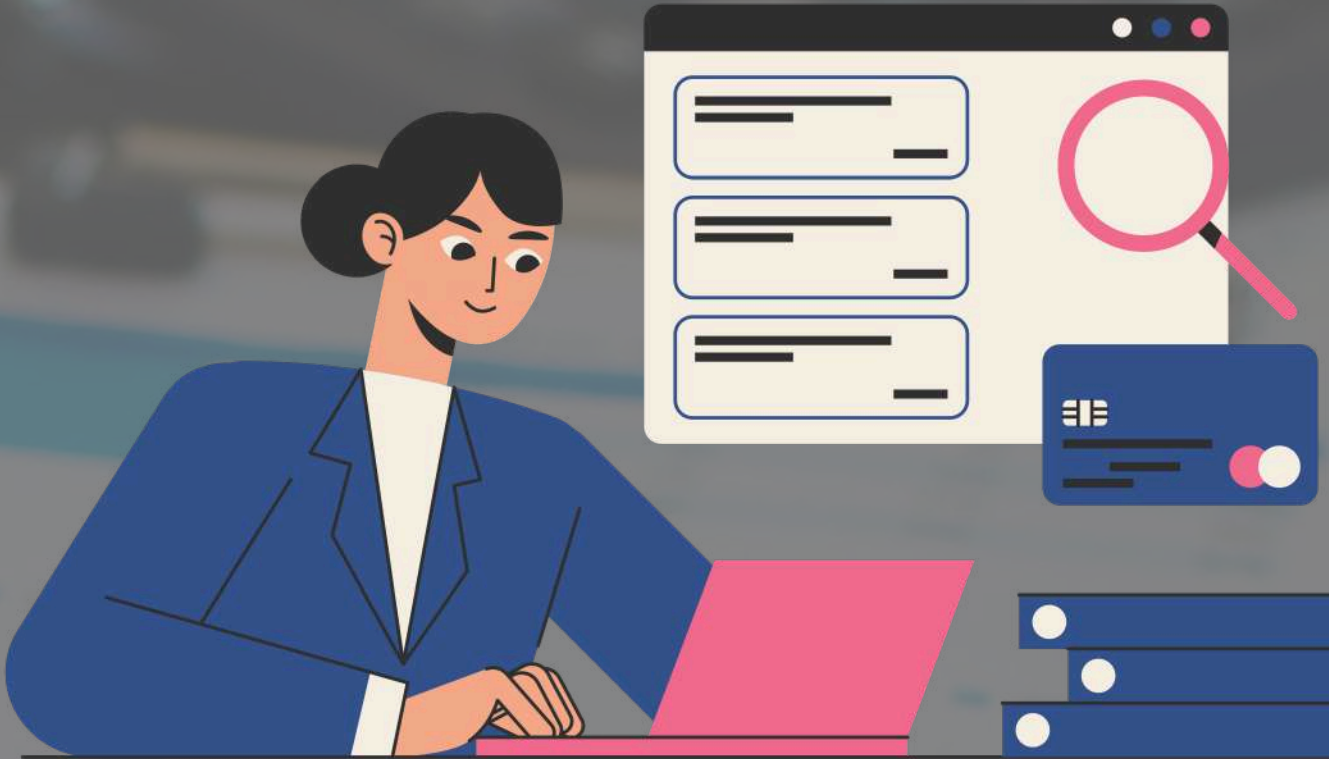


Chart Type	Best Used For
Line Chart	Trends over time
Bar Chart	Comparing categories
Pie Chart	Showing proportions
Scatter Plot	Relationship between variables

Best Practices in Dashboard Design

4. Maintain Simplicity

- Keep the dashboard clean and uncluttered.
- Avoid displaying too much information on one screen.
- Include only the most relevant metrics and visualizations.

Benefits

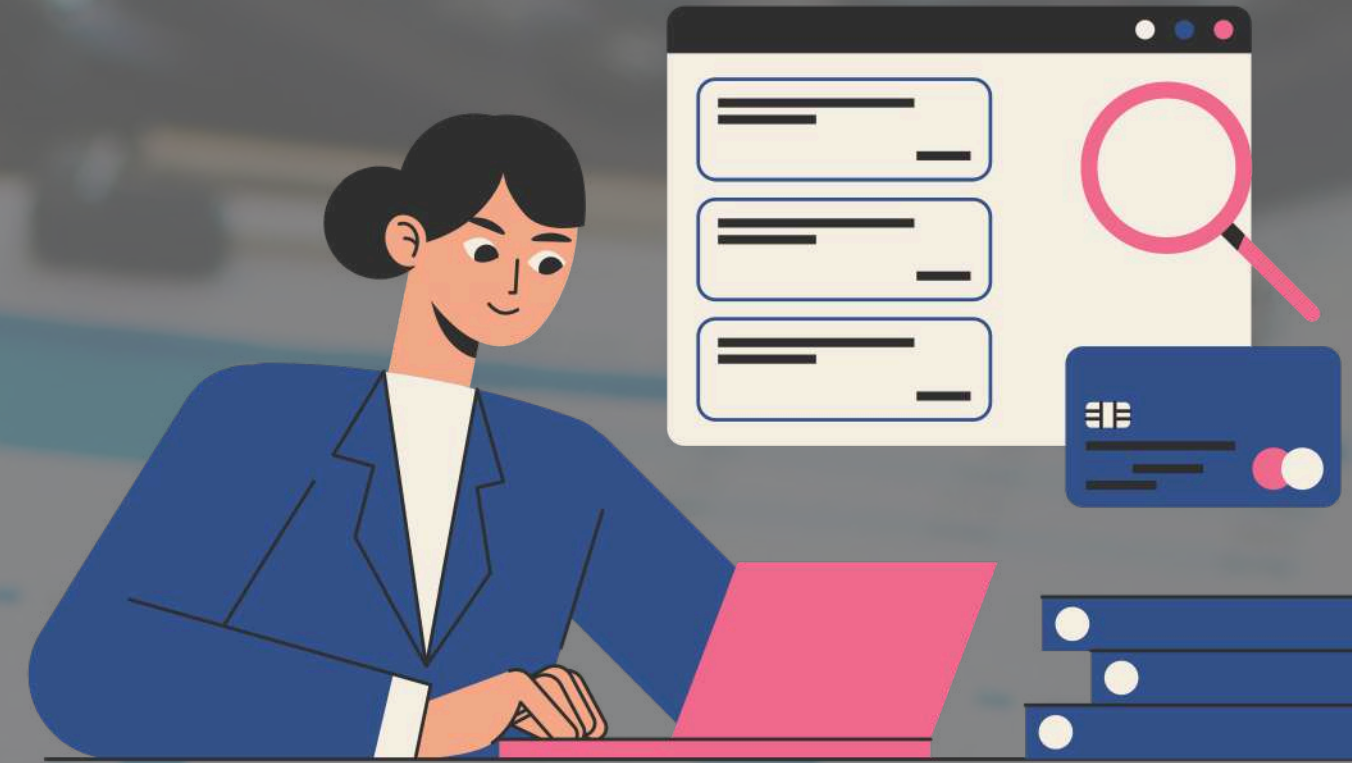
- Better readability.
- Faster understanding.
- Reduced user confusion.

5. Use a Consistent Layout and Design

- Maintain uniform colors, fonts, labels, and chart styles throughout the dashboard.
- Consistency makes the dashboard look professional and easier to use.

Guidelines

- Use the same color scheme.
- Keep similar charts aligned.
- Use clear and consistent labels.



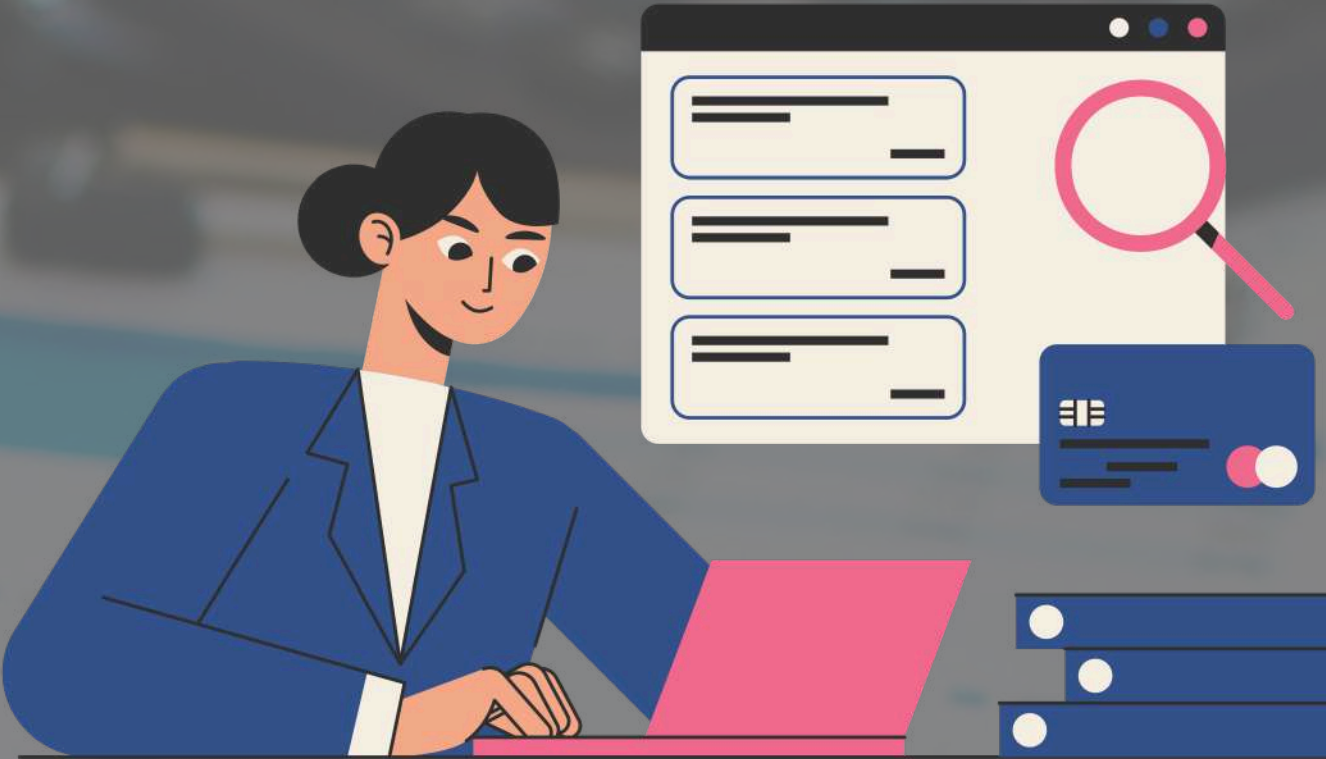
Best Practices in Dashboard Design

6. Highlight Key Metrics (KPIs)

- Important Key Performance Indicators (KPIs) should be clearly visible.
- KPIs help users quickly evaluate business performance.

Examples of KPIs

- Total Revenue
- Total Sales
- Active Users
- Customer Satisfaction Score
- Profit Margin



7. Add Conditional Formatting

- Conditional formatting uses colors, symbols, or icons to highlight important data changes.
- It helps users identify positive and negative performance instantly.

Condition	Display Format
Profit Above Target	Green
Profit Below Target	Red
High Sales Growth	Upward Arrow (↑)
Sales Decline	Downward Arrow (↓)

How Business Reports Help an Organization

Introduction

- A business report is a formal document that presents important information, data, and analysis about different business operations.
- These reports help managers, executives, and stakeholders understand how the organization is performing.
- The main goal of business reports is to support planning, decision-making, and problem-solving.
- They provide accurate and structured information for business growth and improvement.

How Business Reports Help Organizations

1. Improved Decision Making

- Business reports provide accurate and up-to-date information.
- They help managers and leaders make better and faster decisions.
- Reports reduce uncertainty by providing factual data and analysis.

Example

- Sales reports help managers decide which products need more promotion.



How Business Reports Help an Organization

2. Tracking Performance

- Reports help organizations monitor Key Performance Indicators (KPIs).
- They provide insights into business performance over time.

Examples of KPIs

- Revenue
- Profit
- Customer Satisfaction
- Employee Productivity
- Sales Growth

Benefit

- Helps determine whether business goals are being achieved.

3. Identifying Problems Early

- Reports help detect issues before they become major problems.
- They highlight areas where performance is declining.

Benefit

- Enables timely corrective actions.



How Business Reports Help an Organization

4. Planning and Forecasting

- Reports provide a clear view of past and present performance.
- This information helps businesses predict future trends and plan effectively.

Examples

- Forecasting future sales.
- Planning inventory requirements.
- Budget preparation.

Benefit

- Supports strategic business planning.

5. Better Communication

- Reports provide a structured way to communicate information.
- They help share important data across departments and with stakeholders.

Examples

- Management reports
- Financial reports
- Project status reports



How Business Reports Help an Organization

6. Compliance and Documentation

- Business reports serve as official records of business activities.
- They help organizations comply with legal and regulatory requirements.

Examples

- Audit reports
- Financial statements
- Tax reports
- Compliance reports

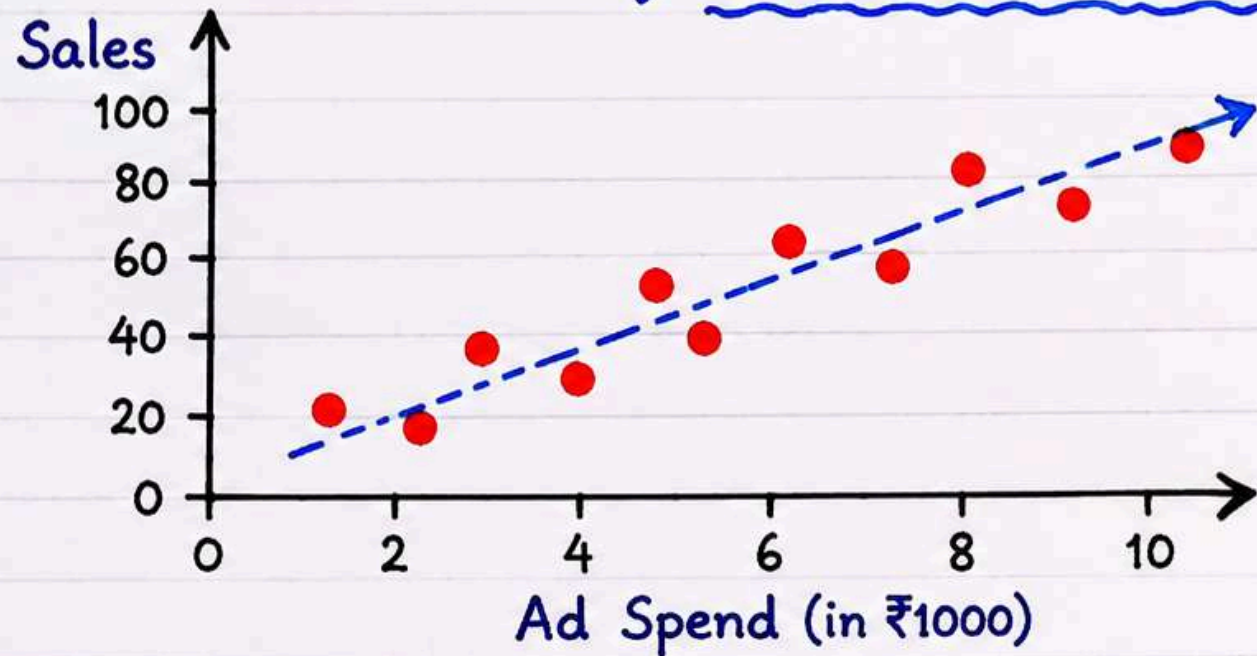
Benefit

- Supports audits, legal reviews, and financial inspections.



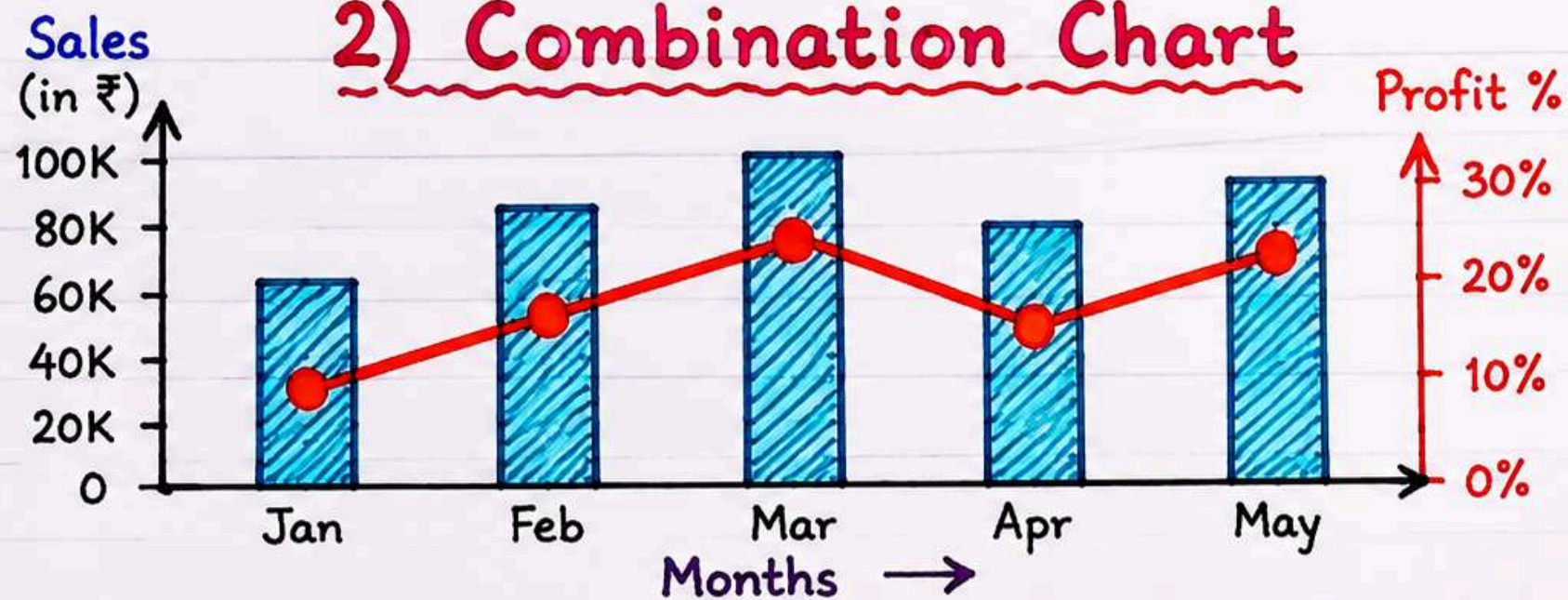
Scatter Chart and Combination Chart

1) Scatter Chart



- Shows relationship between two numerical values.
- Each point represents a pair of values.
- Helps find patterns, trends and correlation.

2) Combination Chart



■ Sales : Bars —●— Profit % : Line

- Uses two or more chart types in a single graph.
- Helps compare different metrics together.
- Bars show quantity, Line shows trend.

Data Integration

Definition

- Data Integration is the process of combining data from different sources into a single, unified view.
- Organizations often collect data from multiple systems such as Sales, Finance, Marketing, and Human Resources.
- Each system stores data in its own format, making analysis difficult.
- Data Integration brings all scattered data together into a centralized system such as a Data Warehouse.
- This allows organizations to analyze data more efficiently and effectively.
- The main goal of Data Integration is to ensure that data is complete, accurate, and consistent across all systems.



Methods of Data Integration

1. Manual Data Integration

Data Integration

Definition

- Manual Data Integration involves collecting data from different sources manually and combining it using tools such as Microsoft Excel or basic scripts.
- Users gather data from multiple systems and merge it themselves.

Characteristics

- Simple to implement.
- Suitable for small-scale projects.
- Useful for one-time data integration tasks.

Limitations

- Time-consuming.
- Error-prone.
- Difficult to manage large volumes of data.

Example

- Exporting sales data and customer data into Excel and combining them manually.



Data Integration

2. Automated (ETL) Data Integration

Definition

- ETL stands for:
 - Extract
 - Transform
 - Load
- It is an automated and advanced method of integrating data.

ETL Process

Step 1: Extract

- Data is collected from different sources such as databases, files, and applications.

Step 2: Transform

- Data is cleaned, filtered, standardized, and converted into a common format.

Step 3: Load

- The transformed data is loaded into a Data Warehouse or centralized storage system.



Data Integration

Step	Description
Extract	Collect data from various sources
Transform	Clean and convert data into a common format
Load	Store data in a Data Warehouse

Advantages of ETL

- Highly efficient for large-scale data operations.
- Reduces manual effort.
- Improves data accuracy and consistency.
- Supports real-time and large-volume data processing.
- Widely used in Business Intelligence (BI) and Data Warehousing systems.



Binning - How it is used

BINNING – HOW IT IS USED

Binning is the process of grouping continuous data into fixed ranges or “bins”. Instead of showing individual values, we organize them into categories.

- For example, if you have customer ages from 18 to 60, instead of showing each age, you can group them into bins like 18–25, 26–35, 36–45, 46–50.

Example: Customer Ages

Original Data (Continuous Values)

18	20	22	24	27	31	33	37	42	46	50	58	60
----	----	----	----	----	----	----	----	----	----	----	----	----



After Binning (Categorized into Bins)

18–25	26–35	36–45	46–50	51–60
18, 20, 22, 24	27, 31, 33	37, 42	46, 50	58, 60

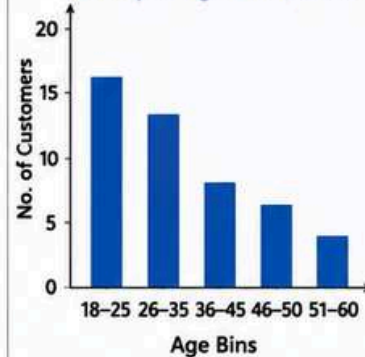
- Binning makes it easier to analyze patterns, spot trends and simplify large datasets.
- In reports, binning improves clarity by converting complex data into easy-to-understand categories.

HOW BINNING IS USED IN REPORT CREATION

1. Histograms & Bar Charts

- Binning is used to create histograms where data is grouped into bins and shown as bars.

Example: Age Distribution



2. Category Creation

- In reports, binning can convert continuous values (like Income) into categories such as:

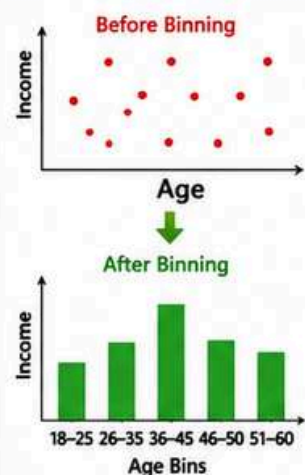
Low Income
(0 – 20000)

Middle Income
(20001 – 50000)

High Income
(50001+)

3. Improved Visualization

- It makes visuals more organized and readable by reducing clutter.



4. Data Segmentation

- It helps in segmenting customers or data points into meaningful groups for marketing, decision-making, etc.



Example

Data (Ages):

[22, 24, 25, 27, 31, 33, 37, 42, 46, 50]



Bin 1
20 – 29
(22, 24, 25, 27)

Bin 2
30 – 39
(31, 33, 37)

Bin 3
40 – 49
(42, 46)

Bin 4
50 – 59
(50)





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